

Perfect Pth Powers

🔖 Tags:

Time Limit: 1 s

Memory Limit: 128 MB

Submission: 79

AC: 34

Score: 99.79

Submit

Codes

AI Code Tips

Description

We say that x is a perfect square if, for some integer b , $x = b^2$. Similarly, x is a perfect cube if, for some integer b , $x = b^3$. More generally, x is a perfect p th power if, for some integer b , $x = b^p$. Given an integer x you are to determine the largest p such that x is a perfect p th power.

Each test case is given by a line of input containing x . The value of x will have magnitude at least 2 and be within the range of a (32-bit) *int* in C, C++, and Java. A line containing 0 follows the last test case.

For each test case, output a line giving the largest integer p such that x is a perfect p th power.

Samples

input Copy

```
17
1073741824
25
0
```

output Copy

```
1
30
2
```

Submit

Codes

HZNUOJ is based on HUSTOJ

--- Maintainer List ---

Server Time: 2025-8-26 15:06:50